



Q1 Sum the given number upto single digit

5 marks

Write a Python code to find the sum of the given number upto single digit

input: 89789

output : 5

Your Code

```
1 n=int(input())
2 while n>=10:
3     s=0
4     while n>0:
5         s=s+(n%10)
6         n=n//10
7     n= s
8 print(n)
```

Input

89789

Output

5

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Q2 Prime Number

5 marks

Find the next prime number

input: 13

output: 17

Your Code

```
1 n=int(input())
2 while True:
3     n=n+1
4     for i in range(2,n):
5         if(n%i==0):
6             break
7     else:
8         print(n)
9         break
```

Input

13

Output

17

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Q3 Swap the first and last digits

5 marks

WAP to swap the first and last digits of the given number.

Input: 23546

Output: 63542

Your Code

```
1 n=int(input())
2 s=str(n)
3 num=s[-1]+s[1:-1]+s[0]
4 print(num)
```

Input

23546

Output

63542

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Q4 Odd Sequence first and Even Sequence next

5 marks

WAP to odd Sequence first and Even Sequence next

Input: 658134

Output: 513684

Your Code

```
1 n = input().strip()
2 odd = []
3 even = []
4
5 for ch in n:
6     d = ord(ch) - 48
7     if d % 2 != 0:
8         odd.append(ch)
9     else:
10        even.append(ch)
11
12 print(''.join(odd + even))
```

Input

658134

Output

513684

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WAP to check the given number is step number or not

A step number is a number in which the difference between every pair of consecutive digits is 1.

For example:

54567 → Yes

$|5-4| = 1$

$|4-5| = 1$

$|5-6| = 1$

$|6-7| = 1$

1235 → No

$|3-5| = 2$

Your Code

```
1 n = input().strip()
2 is_step = True
3 for i in range(len(n) - 1):
4     if abs(int(n[i]) - int(n[i+1])) != 1:
5         is_step = False
6         break
7 if is_step:
8     print("Yes")
9 else:
10    print("No")
```

Input

54567

Output

Yes



Q6 Reverse only the even numbers

5 marks

WAP to reverse only the even numbers

Input: 325649

Output: 345629

Your Code

```
1 n = input().strip()
2 digits = list(n)
3 even = [int(d) for d in digits if int(d) % 2 == 0]
4 even.reverse()
5 idx = 0
6 for i in range(len(digits)):
7     if int(digits[i]) % 2 == 0:
8         digits[i] = str(even[idx])
9         idx += 1
10 print("".join(digits))
```

Input

325649

Output

345629

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WAP to encode the given number

Input: 55553322255

Output: 54322352

Your Code

```
1 n = input().strip()
2 if n == "":
3     print("")
4 else:
5     res = []
6     i = 0
7     while i < len(n):
8         ch = n[i]
9         j = i
10        while j < len(n) and n[j] == ch:
11            j += 1
12        res.append(ch)
13        res.append(str(j - i))
14        i = j
15    print("".join(res))
```

Input

55553322255

Output

54322352

Visible Test Cases

Test Case 1



5.00 / 5 marks

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Reverse Odd Digits

5 marks

WAP to Reverse Odd Digits Only

Input: 816253

Output: 351

Your Code

```
1 n = input().strip()
2
3 odd_digits = [ch for ch in n if int(ch) % 2 == 1]
4 odd_digits.reverse()
5
6 print("".join(odd_digits))
```

Input

816253

Output

351

Visible Test Cases

Test Case 1



5.00 / 5 marks

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WAP to convert every number into the next sequence number.

Input: 89781

Output: 90892

Your Code

```
1 s = input().strip()
2
3 ans = []
4 for ch in s:
5     d = int(ch)
6     ans.append(str((d + 1) % 10))
7
8 print("".join(ans))
```

Input

89781

Output

90892

Visible Test Cases

Test Case 1



5.00 / 5 marks

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Q10 Print the total factor count of a given number

5 marks

WAP to print the total factor count of a given number

Input: 12

Output: 6

Your Code

```
1 n = int(input())
2
3 count = 0
4 for i in range(1, n + 1):
5     if n % i == 0:
6         count += 1
7
8 print(count)
```

Input

12

Output

6

Visible Test Cases

Test Case 1



5.00 / 5 marks

✓ Submitted